

G9-CHECK12-COMPACTNESS-BLOCKED-v1: Step1 partial, Step2 blocked with C_PROVEN

ZFL-CX

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$$\boxed{G7 = FROZEN, G8 = CLOSED/BLOCKED}$$

$$\boxed{\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}, C_J^{neg}\}, \quad T^* < \infty}$$

1 Step1: centers/scales

C_F^{conc} gives $t_n \uparrow T^*$, $x_n, r_n \downarrow 0$ with $\|u(t_n)\|_{L^3(B_{r_n}(x_n))} \geq \varepsilon_0$. C_I^{freq} gives $N_c(t_n) \rightarrow \infty$. But $\boxed{C_I^{freq} \not\Rightarrow r_n = N_c^{-1}}$ identity, and concentration scale vs frequency scale not proved coincident.

Thus: $\boxed{\text{Step1} = \text{POSSIBLE as candidate sequence, NOT closed as intrinsic scale}}$.

Rescaling: $u_n(y, s) = r_n u(x_n + r_n y, t_n + r_n^2 s)$, $s \in (-t_n/r_n^2, (T^* - t_n)/r_n^2)$, window $\rightarrow (-\infty, 0]$ formally.

2 Step2: compactness - THE BOTTLENECK

After scaling:

$$\|u_n(0)\|_{L^3(B_1)} = \|u(t_n)\|_{L^3(B_{r_n})} \geq \varepsilon_0.$$

This is lower bound, excellent for $\bar{u} \neq 0$ IF convergence holds.

But we do NOT have:

$$\|u_n\|_{L^3(B_R \times (-S, 0))} \leq C(R, S), \quad \|u_n\|_{L_t^\infty L_x^3} \leq C.$$

Leray-Hopf bounds $L_t^\infty L^2, L_t^2 \dot{H}^1$ are supercritical under NS scaling $u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t)$. No automatic critical uniform bound.

Pressure: $-\Delta p = \partial_i \partial_j (u_i u_j)$ requires simultaneous control of p_n for compactness.

C_I^{freq} : $\boxed{\text{has structure at } r_n \not\Rightarrow \text{family precompact at } r_n}$. C_J^{neg} : directional regularity loss, no uniform bound on $u_n, \nabla u_n$.

Thus:

$$\boxed{\mathcal{C}_{PROVEN} \not\Rightarrow \text{uniform critical compactness}}$$

$$\boxed{\text{Leray-Hopf} + r_n \downarrow 0 \not\Rightarrow u_n \text{ precompact}}$$

$$\boxed{r_n \rightarrow 0 \not\Rightarrow u_n \rightarrow \bar{u}}$$

3 ZFL verdict

$T^* < \infty \Rightarrow \text{can select } (x_n, t_n, r_n)$ but $\not\Rightarrow u_n \text{ precompact}$.

$G9 \text{ Step1} = \text{PARTIAL/OPEN}$, $G9 \text{ Step2} = \text{BLOCKED}$ with $\mathcal{C}_{PROVEN}$ alone.

This is NOT failure, it's wanted audit result. Naive $T^* < \infty \Rightarrow \bar{u}$ ancient $\Rightarrow$ Liouville $\Rightarrow \perp$ cannot start with five necessary conditions alone.

4 Next micro-audit - one question only

Missing piece:

critical-scale bound producing compactness OR different mechanism giving weak compactness + rigidity without

ZFL rule: DO NOT introduce compactness hypothesis just to make G9 work, then we prove $C_{PROVEN} + hyp \Rightarrow \perp$, not $C_{PROVEN} \Rightarrow \perp$.

Next audit:

$G9\text{-COMPACTNESS-AUTO-AUDIT-v1}$

Question:

Is there automatic suitable-weak / local-energy compactness from $T^* < \infty$ alone, without Type I nor uniform cr

If NO: G9=CLOSED/BLOCKED clean. If YES: new door genuinely different.